

Acoustic Monitoring for Migratory Birds in Lithuania – Land of the Storks

2025 Grant

University of Northern British Columbia

Sunny (Yi-Chin) Tseng

O: 7789852258

3333

University Way

Prince George, British Columbia V2N 4Z9

Sunny (Yi-Chin) Tseng

1228 Arbutus Street

sunnyyctseng@gmail.com

Vancouver, British Columbia V6T3W6

O: 7789852258

Application Form

Eligibility

If you have any questions about eligibility or any other part of the grant process, please feel free to email our committee at grants2025@wildlifeacoustics.com

Organization eligibility*

Eligible organizations are Universities, charities, non-profits or schools. We do not give grants to individuals or government agencies.

My organization is a charity, non-profit, University or school

Researcher and Organization Details

Organization Description*

Please describe the organization applying for the grant

The University of Northern British Columbia (UNBC) is a leading Canadian research university known for its focus on wildlife ecology, conservation biology, and environmental sustainability. Situated in northern British Columbia, UNBC provides strong support for field-based and interdisciplinary research that advances biodiversity monitoring and conservation practices regionally and globally.

Primary Researcher Name*

Sunny Tseng

Primary Researcher Credentials and Official Title

PhD candidate, independent ecologist and data analyst

Primary Researcher Biography*

Please provide a short biography of the primary researcher

Sunny Tseng is a PhD candidate at the University of Northern British Columbia whose research explores how artificial intelligence and bioacoustics can transform bird monitoring and conservation. Originally from Taiwan and now based in Vancouver, Canada, she has conducted bird research across Siberia, Taiwan, Canada, and Lithuania, collecting vocalizations from more than 300 species. Sunny has been awarded an NSERC PhD fellowship, a National Geographic Explorer Grant, and several other research fellowships recognizing her expertise in avian acoustic monitoring and ecological data analysis.

Her PhD research aims to improve the reliability of AI-based bird sound identification by developing species-specific confidence thresholds for BirdNET detections. She also leads the development of **birdnetTools**, an open-source R package that streamlines post-processing and validation of BirdNET outputs, helping bridge the gap between AI technology and practical biodiversity monitoring.

Secondary Researcher Name*

If the primary researcher is unable to complete the project, this person would become the contact for the grant
Modestas Bružas

Secondary Researcher Email*

modestas.bruzas@nerija.lt

Secondary Researcher Phone Number*

+370 60368766

Kaleidoscope Only Grant Application

Is this a Kaleidoscope Only Grant Application?

Please check this only if you are just requesting Kaleidoscope Pro Software. If you wish to apply for hardware as well, please ignore this question.

Training

Have you attended any Wildlife Acoustics trainings?*

Yes

If you've attended any trainings, please list:

I attended the Wildlife Acoustics information session during the 2025 Society of Canadian Ornithologists (SCO-SOC) Conference held in Saskatoon.

Project Details

Project Name*

Name of Project.

Acoustic Monitoring for Migratory Birds in Lithuania – Land of the Storks

Project Location*

Describe the country, region, or nearest city. You may also upload a map or picture.

fig_study_map.jpg This project takes place in **Lithuania**, a Baltic country with a rich mosaic of ecosystems that support over 300 bird species and lie along a major migratory flyway. Fieldwork will occur in two ecologically significant protected areas: **Curonian Spit National Park**, a UNESCO World Heritage Site of dunes, forests, and wetlands that borders the long-established **Ventės Ragas Ornithological Station**, and **Žuvintas Biosphere Reserve**, Lithuania's largest wetland and a UNESCO Biosphere Reserve, home to over 240 bird species,

including globally threatened taxa. Twelve autonomous recording units (ARUs) will be deployed across multiple habitat types at each site, spaced to maximize coverage, with support from local collaborators **Tomas Pikūnas** (Žuvintas manager) and **Modestas Bružas** (Curonian Spit biologist).

Study Subject*

You may choose more than one

bird

frog

Study Subject/Other

If you selected other above, please describe your subject here

Short Project Description*

Please give us a short description of your project. This will be used to help evaluators keep track of your application and if chosen, help us create a good description of your work for the announcement on our website and email.

This project aims to establish Lithuania's first long-term passive acoustic monitoring program for migratory birds, providing baseline biodiversity data across key habitats. Pilot recordings from the 2025 breeding season demonstrated strong potential for large-scale implementation. With support from the Wildlife Acoustics Grant Program, we will expand monitoring in 2026 at two ecologically significant sites—Žuvintas Biosphere Reserve and Curonian Spit National Park—in full collaboration with local partners. The project will generate a comprehensive baseline dataset to support long-term avian monitoring and conservation in Lithuania.

Choose the one that best describes your project*

Monitoring to Inform Species Conservation

If you selected other, please describe here

What is the problem being solved or the question being answered?*

Lithuania is home to one of Europe's most diverse bird communities, with critical stopover and breeding habitats along the Baltic Flyway, including lakes, wetlands, dunes, and forests. Despite this ecological significance, the country currently lacks a systematic, long-term program for monitoring bird biodiversity using passive acoustic methods, and existing documentation of bird vocalizations is limited. This gap constrains the ability of researchers and conservation managers to track species presence, assess population trends, and make informed habitat management decisions.

This project addresses this need by establishing Lithuania's first multi-year passive acoustic monitoring program for migratory birds. Using strategically deployed autonomous recording units, we will document species presence, vocal activity, and temporal patterns across key habitats. The resulting datasets will provide a permanent record of avian biodiversity, support national partners in conservation planning, and contribute open-access data to global initiatives, helping to fill a critical knowledge gap in both local and international biodiversity monitoring efforts.

How do you plan to use bioacoustics to meet the objectives of the project?*

We will use **passive acoustic monitoring** with autonomous recording units (ARUs) to document bird presence and vocal activity across diverse habitats in Lithuania's **Žuvintas Biosphere Reserve** and **Curonian Spit**

National Park. Recordings collected during the 2026 breeding season will be analyzed using **AI-based detection tools** like BirdNET, combined with manual verification to ensure accurate species identification. This approach will allow us to quantify temporal patterns of vocal activity, characterize community composition, and create a permanent, shareable dataset. The resulting data will support conservation management, provide national partners with actionable insights, and contribute to open-access repositories, establishing Lithuania's first long-term acoustic monitoring program for migratory birds.

Besides publishing your results and presenting at conferences, how will the results be used?*

Beyond publishing results and presenting at conferences, the project will engage a wide range of audiences through targeted outreach. Internationally, findings will be shared via **National Geographic**, including an English article and a Mandarin version to reach readers in Taiwan and other Mandarin-speaking regions. Locally, Sunny Tseng will return to Lithuania in 2026 to conduct fieldwork and host workshops and presentations in collaboration with the **Taiwanese Representative Office**, engaging the **Lithuanian Ornithological Society**, universities such as Vilnius University and Vytautas Magnus University, and other conservation stakeholders. These activities will demonstrate acoustic monitoring techniques, promote long-term ecological research, and foster international collaboration. By connecting researchers, practitioners, and the public across countries, the project will enhance appreciation for biodiversity and the innovative tools used to protect it.

How does this project impact or expand wildlife conservation or research?*

This project advances wildlife conservation and research by establishing Lithuania's first long-term passive acoustic monitoring program for migratory birds. Using non-invasive ARUs and AI-based analysis, it will document species presence, vocal activity, and community dynamics across diverse habitats. The resulting baseline dataset will inform habitat management and conservation planning, while training local partners and contributing data to open-access platforms like **Xeno-Canto**. By combining cutting-edge technology with long-term monitoring, the project strengthens understanding of avian biodiversity and demonstrates the value of scalable, replicable bioacoustic approaches for conservation worldwide.

Please add any additional information about your project (optional)

Have you applied for this grant before?*

(Your answer will not negatively impact your application.)

No

Eligible Products

Please indicate which products you're requesting.

Recorders

Song Meter Micro 2

Accessories for Mini and Mini Bat

[Unanswered]

Echo Meter Touch 2 Software

Product Selection

Specify the quantities you need of each product.

Song Meter Mini 2 AA \$499

[Unanswered]

Song Meter Mini 2 Li-Ion \$599

[Unanswered]

Song Meter Mini Bat 2 AA \$749

Song Meter Mini 2 Bat Li-Ion \$849

Song Meter Micro 2 \$149

26

Acoustic Stub Mic for SongMeter Mini/Mini Bat \$75

Li-ion Batteries (pair) \$30

[Unanswered]

Ultrasonic Calibrator \$199

Replacement Windscreens (pair) \$15

[Unanswered]

Echo Meter Touch 2 Standard for Android \$179

Echo Meter Touch 2 PRO for Android \$349

Number of Kaleidoscope Pro Licenses Requested \$399

One License represents one user for one year of use

Total Value of Product(s) Requested:*

Follow this link to our webstore to find the value of each product. Type in the total value below. You may request up to \$4000 worth of equipment and software

\$3,874.00

Methods

Overview of your methods*

We will establish Lithuania's first long-term passive acoustic monitoring program for migratory birds at **Curonian Spit National Park** and **Žuvintas Biosphere Reserve**. **Twelve autonomous recording units (ARUs)** will be deployed at each site across multiple habitat types, spaced to minimize overlap and maximize spatial coverage. ARUs will record continuously during the 2026 breeding season, capturing both diurnal and nocturnal vocalizations. Recordings will be processed using automated species recognition software (e.g., BirdNET) to identify species and generate detection data. Data will be analyzed to assess species richness, temporal activity patterns, and habitat associations, providing a baseline dataset for long-term monitoring. Local collaborators will assist with deployment, maintenance, and ground-truthing to ensure data quality.

Describe in detail how the recorders will be deployed*

Such as number of recorders, where they will be deployed, when they will record, etc

Recorders will be deployed at 12 sites each in Curonian Spit National Park and Žuvintas Biosphere Reserve, for a total of 24 autonomous recording units (ARUs). At each site, ARUs will be mounted approximately 1.5 meters above the ground on trees, oriented north to avoid direct sunlight and minimize weather exposure. Recordings will be collected continuously during the main field season (May–July 2026) using a duty cycle of 1 minute on, 4 minutes off, providing broad temporal coverage while maximizing battery life and storage. This deployment design ensures representative sampling across habitats, captures daily activity patterns, and facilitates long-term comparisons of avian vocal activity.

Describe in detail how the recordings will be analyzed*

What software will be used and why? Who will do the analysis?

Recordings will be analyzed using BirdNET, chosen for its extensive coverage of international bird species, strong performance with European avifauna, and proven reliability in research applications, including the primary investigator's PhD work. BirdNET's AI-based algorithms will automatically detect and classify bird vocalizations, providing an efficient and standardized approach to processing large volumes of acoustic data. Key species identifications will undergo manual verification to ensure accuracy. The analysis will be conducted by Sunny Tseng, with support from trained collaborators, ensuring consistent, high-quality processing and the generation of reliable datasets for subsequent ecological analyses and conservation planning.

Budget

Budget*

Provide a budget that clearly indicates the complete project costs, including the amount of equipment being requested from Wildlife Acoustics. Please use US dollars. You may also upload a file if you prefer.

The total project budget is approximately **\$13,000 USD**. We request **\$4,000** from the Wildlife Acoustics Grant Program to cover the cost of eight autonomous recording units (ARUs) with microphones and accessories. Additional project costs, including field supplies and equipment (mounting poles, weatherproof housings, batteries, solar panels, GPS units), will total around **\$6,000**, while travel and transportation to field sites are estimated at **\$3,000**. Remaining expenses will be supported by other funding sources, including the National Geographic Early Career Grant and the Pairi Daiza Foundation Grant.

What else, other than Wildlife Acoustics equipment, is needed to ensure project success?*

Include resources, other equipment, permits, etc

In addition to Wildlife Acoustics equipment, the project will require:

- **Field supplies and logistics:** batteries, SD cards, and weatherproof storage for fieldwork.
- **Permits and approvals:** research and access permits from the **Curonian Spit National Park** and **Žuvintas Biosphere Reserve**, and ethical approval for non-invasive wildlife monitoring as required by Lithuanian authorities.
- **Local support:** collaboration with site managers and biologists for deployment, site selection, and field verification, ensuring accurate placement of ARUs and minimizing disturbance to sensitive habitats.
- **Transportation:** vehicles or boats to access remote sites, particularly in wetland and lagoon areas, for deployment and maintenance.

Other Funding*

What other resources and funding are required for the project?

This project will be supported through a combination of funding sources. **National Geographic Early Career Grant** and **Pairi Daiza Foundation Grant** will contribute to fieldwork logistics, travel, and local collaboration costs. The **Wildlife Acoustics Grant Program** will provide essential ARU equipment for passive acoustic monitoring. Together, these resources ensure the successful implementation of Lithuania's first long-term migratory bird monitoring program.

Is all other funding secured? If not, why?*

Not all other funding is fully secured at this time. National Geographic supported the 2025 fieldwork, and discussions are ongoing to continue funding for 2026. The outcome of the Pairi Daiza Foundation Grant will be announced later this year. All funding is expected to be confirmed prior to the 2026 field season.

If this grant is not awarded, will the project proceed? If so, how?*

Yes, the project will still proceed if the fieldwork logistics are funded by **National Geographic** and the **Pairi Daiza Foundation**. Equipment typically supported by the **Wildlife Acoustics Grant** could be borrowed from organizations such as **Birds Canada** or **Environment and Climate Change Canada**. However, this approach is not ideal, as borrowed devices may differ in model (e.g., AudioMoth, Wildlife Acoustics, or Frontier), introducing variability and potential machine-specific bias in the recordings.

Project Timeline

Project Timeline*

What are the phases of the project? When are they expected to start and end? Include a summary timeline of the project with dates and phases indicated.

Screenshot 2025-10-30 140010.png The project will begin with securing permits, coordinating logistics with local collaborators, and preparing equipment. The main field season runs from **May to July 2026**, followed by systematic data processing, species verification, and preliminary analyses (see attached timeline).

This timeline ensures a **structured workflow** from preparation through data collection, analysis, and dissemination, supporting the establishment of Lithuania's first long-term acoustic monitoring program for migratory birds.

Estimated Project Start Date*

You must plan to start the project within one year.

05/01/2026

Estimated Project End Date*

12/31/2026

Tax Exemption Details

Please check one box*

You will be directed to the appropriate next question below

My country does not provide tax exempt documents

Tax Certification

Tax certification*

If your country does not provide tax exempt documents, please check the box below certifying that, if you win the grant, your organization will pay any applicable taxes and duties.

I agree to pay for all applicable taxes and duties

Referral letter #2 - this second letter is optional

Send an email to a second referral (optional)

- Enter the email address of the individual from whom you are requesting a letter of recommendation. We strongly suggest that you confirm that the individual has agreed to write your letter.
- Click "Compose Email" and compose an email to your reference requesting that he/she write a letter of recommendation for you. Click on "send". You may copy the example message below to paste into the body of your email:

'Hello! I am applying for a grant through Wildlife Acoustics. I am writing you today to request that you provide a Letter of Recommendation. You will receive a separate email that contains a link to the reference form. Please let me know when you've completed it. Thank you for your time.'

- The recommender will then receive your email and an email from Wildlife Acoustics with a link to the reference form.

Once this email is sent, you can continue completing and then submit your application while this

letter is pending. Your contact will submit their letter on their own directly in our system. However, only applications completed by the deadline, including letters of recommendation, will be considered by our committee.

Referral Letter #1 - this is required

Send an email to your reference*

- Enter the email address of the individual from whom you are requesting a letter of recommendation. We strongly suggest that you confirm that the individual has agreed to write your letter.
- Click "Compose Email" and compose an email to your reference requesting that he/she write a letter of recommendation for you. Click on "send". You may copy the example message below to paste into the body of your email:

‘Hello! I am applying for a grant through Wildlife Acoustics. I am writing you today to request that you provide a Letter of Recommendation. You will receive a separate email that contains a link to the reference form. Please let me know when you've completed it. Thank you for your time.’

- The recommender will then receive your email and an email from Wildlife Acoustics with a link to the reference form.

Once this email is sent, you can continue completing and then submit your application while this letter is pending. Your contact will submit their letter on their own directly in our system. However, only applications completed by the deadline, including letters of recommendation, will be considered by our committee.

ken.otter@unbc.ca

Terms

If awarded this grant, I agree to:*

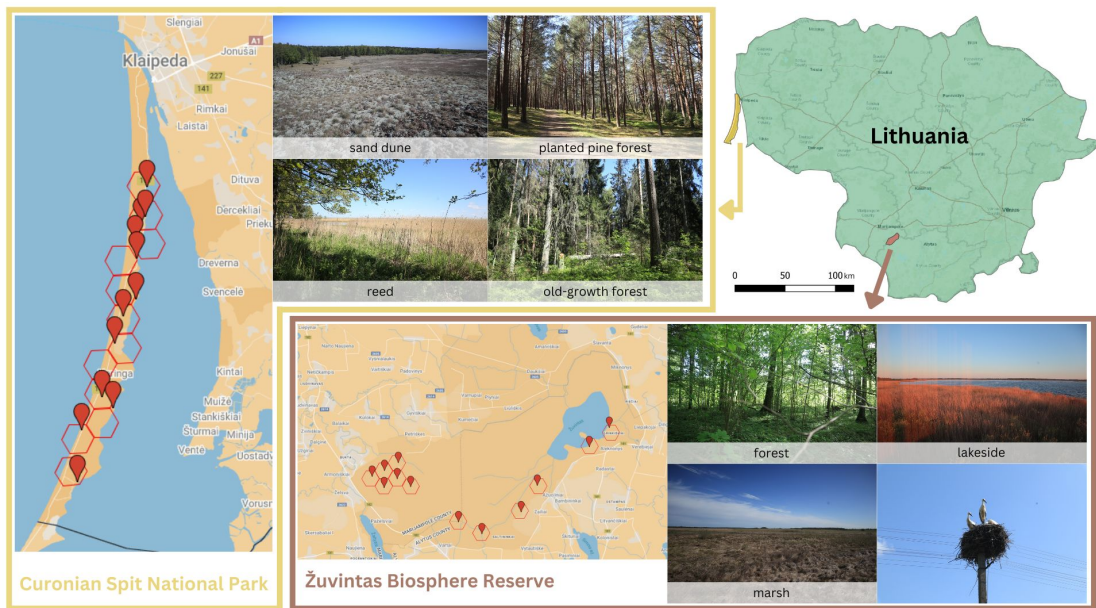
- Pay all import duties and taxes (if applicable) on products shipped outside the US
- Sign a Memo of Understanding with Wildlife Acoustics
- Provide a photo or short video of myself
- Start my project within one year
- Submit 3 interim reports as agreed in the MOU
- Acknowledge the support of Wildlife Acoustics, Inc in any presentations, meetings, publications and materials resulting from the project
- Submit a final report within two months of completing the project or phase of the project for a total of 4 reports

Yes, I agree

File Attachment Summary

Applicant File Uploads

- fig_study_map.jpg
- Screenshot 2025-10-30 140010.png



Work plan.....

The early months of the project are devoted to securing permits with local collaborators, coordinating logistics, and preparing equipment. The main field season runs from May to July, followed by systematic data processing, species verification, and preliminary analyses. The detailed timeline is outlined below:

Period	Activities	Expected outcomes
Jan–Apr 2026	Confirm research permits, coordinate logistics with local partners, calibrate ARUs	All permissions and logistics finalized
May–Jul 2026	Install and monitor ARUs; conduct targeted recordings at both sites	Successful collection of acoustic data
Aug–Oct 2026	Classify recordings using BirdNET; manual verification of key species	Preliminary biodiversity summary
Nov–Dec 2026	Share data with local collaborators, upload species recordings to Xeno-Canto, prepare final report	Final dataset and report submitted to collaborators